

The effects of a mutation in an intrinsically disordered protein propagates through hydrophobic blob interactions

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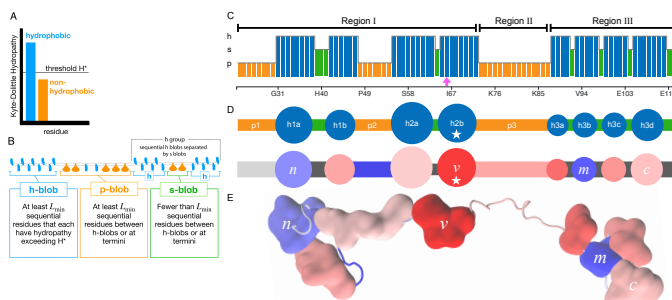


Abstract

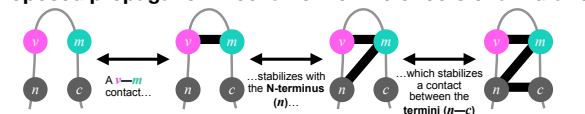
Intrinsically disordered proteins (IDPs) sample a broad distribution of conformational states that comprise the conformational ensemble. Single mutations in IDPs often shift the populations of conformational states, thus altering the global properties of the protein. The effects of charge-altering mutations can be rationalized from changes in electrostatic interactions. However, chemically conservative mutations present a greater challenge because the substituted residue has physicochemical properties similar to those of the original residue. Chemically conservative mutations have been shown to alter interactions between regions distant from the mutation site, affecting global properties and protein function, but the mechanisms by which the effects of these mutations propagate to distal regions of the sequence have not yet been elucidated. Here, we use physicochemically defined sequence regions ("blobs") to classify contacts and distinguish direct associations between the mutation-containing blob and terminal blobs from associations mediated by an intermediate blob. We apply this approach to seven variants of the disordered Brain-Derived Neurotrophic Factor (BDNF) prodomain to test a series of chemically conservative perturbations at the same position. Using 128 μ s of all-atom explicit-solvent enhanced molecular dynamics simulations for each variant, we demonstrate that contacts involving the mutation-containing blob are associated with contacts between intermediate and distal regions of the sequence for most variants, supporting an intermediate-mediated pathway for propagation of the effects of the mutation. This approach provides a framework for identifying interaction pathways through which the effects of a mutation can propagate across the sequence of a long IDP.

Background

- IDPs adopt many conformations that make up the conformational ensemble.
- Over 20% of disease-causing mutations** occur in intrinsically disordered regions [1], and even **chemically conservative mutations** can shift IDP ensembles.
- The **Val66Met mutation in the disordered BDNF prodomain** highlights this by altering its compactness, secondary structure, and **contacts with distal regions** of the sequence [2, 3].
- Our enhanced molecular dynamics simulations, combined with Blobulation (Fig 1A-C), revealed that the mutation-containing Variant blob (v blob in Fig 1D-E) **affects interactions with the distal h3b blob**, and is accompanied by altered interactions with the N-terminal blob (n).
- It remained unclear whether the altered interactions between the v blob and h3b propagated to both termini and whether this propagation pathway involving the h3b is specific to the Met66 variant, or a general pathway for other chemically conservative mutations.
- Here, we present a blob-based contact classification approach to determine whether the effects of the mutation-containing blob **propagate directly to the terminal blobs or indirectly through h3b**, a candidate mediator blob (m).



Proposed propagation mechanism of the effects of a mutation



Approach

- Simulate chemically conservative mutations** (F, M, V, L, A, Y, I) at site 66 of the BDNF prodomain sequence using temperature-replica exchange molecular dynamics for 2 μ s each.
- Apply Blobulation** to reduce dimensionality and identify contacts.
- Classify conformational states** by the presence or absence of contacts that connect the mutation-containing blob and distal regions of the sequence.
- Test interaction pathways** to determine whether the mutation-containing blob is associated with distal contacts directly or through an intermediate blob.

Global properties of the BDNF prodomain are affected by a chemically conservative mutation

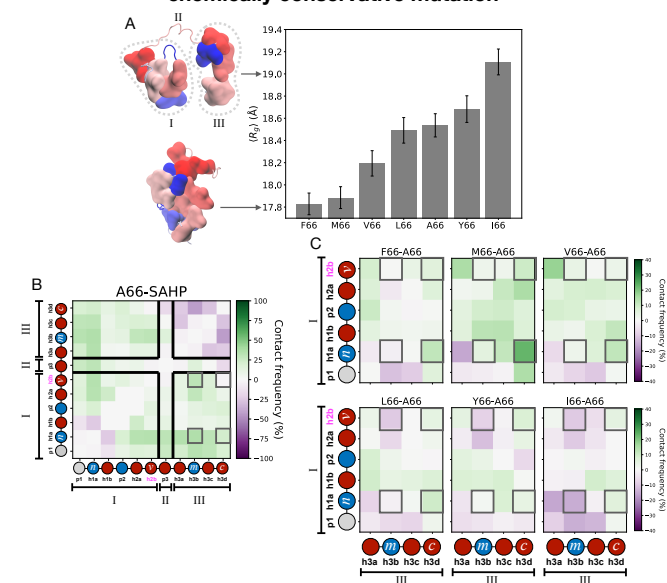


Figure 2: (A) Average radius of gyration ($\langle R_g \rangle$) for all BDNF prodomain variants with VMD images of the IDP at varying R_g values colored by NCPR. **(B)** The difference in blob contact frequency between the A66 variant and the self-avoiding heteropolymer (SAHP). **(C)** The difference in blob contact frequency between all of the variants and A66.

The compactness of the BDNF prodomain is correlated to terminal blob contacts

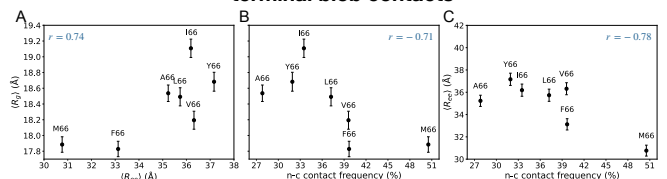


Figure 3: (A) Average radius of gyration, (R_g), vs. the average end-to-end distance, (R_{ee}) and **(B)** the frequency of the N- and C-terminal ($n-c$) blob contacts. **(C)** $\langle R_{nc} \rangle$ vs. the frequency of the $n-c$ contact. Pearson's correlation coefficients are labeled on the plots shown in blue. Error bars represent the standard error of the mean of the y-axis variable.

Terminal blob contacts of the BDNF prodomain are generally associated with contacts involving a mediator blob

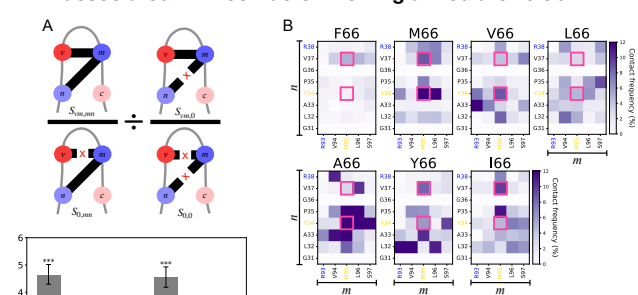


Figure 4: (A) Odds ratios (OR) between $v-m$ and $m-n$ contacts for all BDNF prodomain sequences. These contacts are positively associated with each other (OR > 1). **(B)** Side-chain contacts between residues in S_{m-m} , showing only contacts between the m and n blobs.

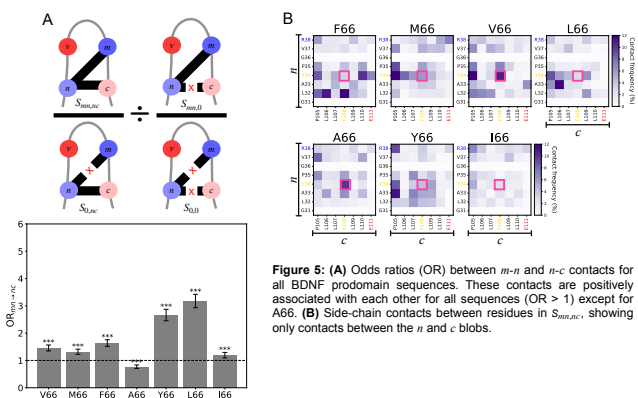


Figure 5: (A) Odds ratios (OR) between $m-n$ and $n-c$ contacts for all BDNF prodomain sequences. These contacts are positively associated with each other (OR > 1) except for A66. **(B)** Side-chain contacts between residues in S_{m-m} , showing only contacts between the n and c blobs.

Summary

- Chemically conservative mutations at the same sequence position shift contact states within the conformational ensemble of the BDNF prodomain.
- Blobulation and contact classification reveal interaction pathways connecting the mutation-containing region to distal regions.
- For most variants, mutation-associated contacts are **coupled to distal contacts through the Mediator blob**.
- Mutations shift distal side-chain interaction preferences
- Our approach provides a **framework for identifying how mutation-associated changes propagate through specific inter-blob contacts** within the conformational ensemble of an IDP.

References

- V. Vucic and L. M. Iakoucheva. *Mol Biosyst.*, 2012
- R. Lohia, R. Salari, and G. Brannigan. *PLoS Comput. Bio.*, 2019.
- A. Anastasia et al. *Nat Commun.*, 2013
- R. Lohia, M.E.B. Hansen, and G. Brannigan. *PNAS*, 2022.
- C. Pitman et al. *JCIM*, 2026

Acknowledgements

- Rutgers Office of Advanced Research and Computing (OARC)
- NRT, NSF DGE 2152059

Research Question

How does a chemically conservative mutation affect distal regions in the intrinsically disordered BDNF prodomain sequence?